

Brazilian Synchrotron Light Laboratory

Gabriel Rezende

Overview of ID tracking methods

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1 Introduction

This document presents a detailed overview of insertion devices tracking methods. The first section is an introduction to the Ellaume formalism, derivations and predicted effects on the beam. The second section presents the tracking method based on Runge-Kutta integration and compares results with those obtained from the Ellaume formalism. Finally, last section presents an symplectic integration method proposed by Wu, Forest and Robin and also compares the results with the two first methods. The main goal of this document is to provide a complete overview of the methods used to track insertion devices in the SIRIUS storage ring and to compare their results.

2 Ellaume formalism

2.1 Equations of motion

Using SIRIUS coordinate system, the equations of motion for a particle in a magnetic field can be expressed as:

$$\frac{d\vec{p}}{dt} = q\vec{v} \times \vec{B} \quad (1)$$

$\vec{p}$ is the momentum of the electron, $q = -e$ is the electron's charge, $\vec{v}$ is its velocity, and $\vec{B}$ is the magnetic field along the electron's path. The velocity can be written as:

$$\vec{v} = \frac{dx}{dt}\hat{x} + \frac{dy}{dt}\hat{y} + \frac{ds}{dt}\hat{s} = \left(\frac{ds}{dt}\right) \left(\frac{dx}{ds}\hat{x} + \frac{dy}{ds}\hat{y} + \hat{s}\right) \quad (2)$$

defining

$$x' := \frac{dx}{ds}, \quad y' := \frac{dy}{ds} \quad (3)$$

and assuming the electron is ultra-relativistic, we have:

$$c^2 = \vec{v}^2 = \left(\frac{ds}{dt}\right)^2 (x'^2 + y'^2 + 1) \quad (4)$$

then, it is found the following expression for ds/dt :

$$\frac{ds}{dt} = \frac{c}{\sqrt{x'^2 + y'^2 + 1}} \quad (5)$$

which gives the expression for the velocity:

$$\vec{v} = \frac{c}{\sqrt{x'^2 + y'^2 + 1}} (x'\hat{x} + y'\hat{y} + \hat{s}) \quad (6)$$

Substituting this expression into the Lorentz force equation, we have:

$$\frac{d\vec{p}}{dt} = \frac{-ec}{\sqrt{x'^2 + y'^2 + 1}} [(y'B_s - B_y)\hat{x} - (x'B_s - B_x)\hat{y} + (x'By - y'Bx)\hat{s}] \quad (7)$$

Keeping the result of equation (7), now let's express the momentum of a relativistic electron:

$$\vec{p} = \gamma m_e \vec{v} = \frac{\gamma m_e c}{\sqrt{x'^2 + y'^2 + 1}} (x'\hat{x} + y'\hat{y} + \hat{s}) \quad (8)$$

and it's time derivative:

$$\frac{d\vec{p}}{dt} = \frac{ds}{dt} \frac{d\vec{p}}{ds} = \frac{\gamma m_e c^2}{\sqrt{x'^2 + y'^2 + 1}} \left[-(x'^2 + y'^2 + 1)^{-3/2} (x'x'' + y'y'') (x'\hat{x} + y'\hat{y} + \hat{s}) + \frac{x''\hat{x} + y''\hat{y}}{\sqrt{x'^2 + y'^2 + 1}} \right] \quad (9)$$

Then, using the magnetic rigidity $R = \gamma m_e c / e$ and equating the $\hat{s}$ component of equations (7) and (9), we have:

$$\frac{x'x'' + y'y''}{(x'^2 + y'^2 + 1)^{3/2}} = \frac{1}{R} (x'B_y - y'B_x) \quad (10)$$

substituting equation above into equation (9), the remaining two vector components are related by

$$-\frac{1}{R} (x'B_y - y'B_x) (x'\hat{x} + y'\hat{y}) + \frac{x''\hat{x} + y''\hat{y}}{\sqrt{x'^2 + y'^2 + 1}} = -\frac{1}{R} [(y'B_s - B_y)\hat{x} + (x'B_s - B_x)\hat{y}] \quad (11)$$

Solving this equation for x'' and y'' , we have the following equations of motion:

$$\begin{aligned} x'' &= -\frac{\sqrt{x'^2 + y'^2 + 1}}{R} [x'y'B_x - (x'^2 + 1)B_y + y'B_s] \\ y'' &= -\frac{\sqrt{x'^2 + y'^2 + 1}}{R} [(y'^2 + 1)B_x - x'y'B_y - x'B_s] \end{aligned} \quad (12)$$

2.2 First-order solutions to the equations of motion

Assuming the transverse velocity of the electrons is zero, the equations of motion reduce to:

$$\begin{aligned} x'' &= \frac{1}{R} B_y \\ y'' &= -\frac{1}{R} B_x \end{aligned} \quad (13)$$

The integration of equation (13) leads to:

$$\begin{aligned} x'(s) &= \frac{1}{R} \int_{-\infty}^s B_y(0, 0, s') ds' \\ y'(s) &= -\frac{1}{R} \int_{-\infty}^s B_x(0, 0, s') ds' \\ x(s) &= \frac{1}{R} \int_{-\infty}^s \int_{-\infty}^{s'} B_y(0, 0, s'') ds'' ds' \\ y(s) &= -\frac{1}{R} \int_{-\infty}^s \int_{-\infty}^{s'} B_x(0, 0, s'') ds'' ds' \end{aligned} \quad (14)$$

These are the so called field integrals. These equations are a good approximation for elements like quadrupoles, sextupoles and correctors, but for a well designed insertion device, the field integrals are close to zero, and thus the first-order solutions are not enough to describe the electron's motion. In this case, it is necessary to consider the second-order solutions to the equations of motion.

2.3 Second-order solutions to the equations of motion

3 Conclusion

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